

Determining fast-charge urban electric bus services:

(O. Battaïa, A. Dolgui, N. Guschinsky, M. Y. Kovalyov)

Introduction to the problem

- **Air Pollution:**

- In 2000s there were 960,000 premature deaths
- By 2030 → nearly 3.1 million projected deaths

- **Electric Bus Adoption**

- Progressive laws requiring low-emission fleets → 100% by 2025
- Growth of the new electric bus market → **+18.4%**

- **Research Context**

- Urban electric buses need efficient charging networks → **Where should charging stations be placed for maximum efficiency?**
- Many studies address only **one aspect** of the problem

Introduction to the problem

- **2 Models:**

- *Previous model:*

- Handling multiple aspects, but **non-linear** and **slow** to solve

- *Current model (Opt-Fast-Char2):*

- **Linear model** → Practical, faster, handling charging stops
 - Number of feasible charging spots is small

- **Can optimization models truly guide real-world city planning?**

- **Key Assumptions in Opt-Fast-Char2:**

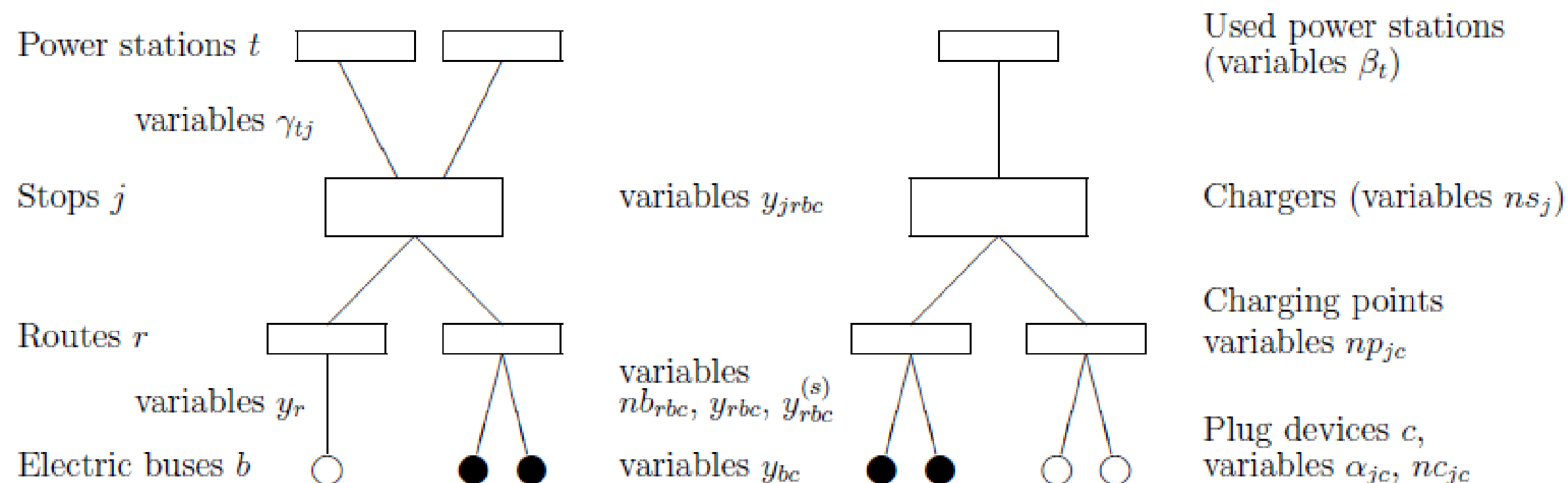
- At most one charge between two terminals
 - Chargers accessible only from one side of the road
 - Objective reduces unused facilities and conventional buses.

System Overview

- **Urban bus network:**
 - Multiple routes → each with a **depot** and a **cycle of stops**
 - Served by electric **and** conventional buses
- **Electric bus diversity:**
 - Several types → Different **passenger capacity** and **charging characteristics**
 - Multiple charging technologies per type
- **Charging infrastructure:**
 - Charging points grouped into **charging facilities**
 - Facilities powered by **electric power stations**
 - Each power station → max 1 facility
 - Each facility → several stations

System overview

- **Bus charging process:**
 - Plugged into charging point via charging plug
 - **Parallel charging** possible if delays occur (at stops)
- **Daily bus cycle:**
 - Start **fully** charged at depot → fixed route → return to depot
 - **Overnight charging** at depots
 - All depots are **mandatory charging stops**



ILP model definition

Instance

- **Public transport system**
 - Real data enriched with self-constructed data
 - Introducing **feasible scenarios** for locations of charging facilities
 - Replacing conventional buses with **electric buses** → maintaining passenger capacity
 - Major limiting factors:
 - **Capital & Variable** costs
 - Supplied & consumed **electric power**
- **Graph representation**
 - **Nodes** representing depots and stops
 - **Edges** representing connections between stops

Variables

Quantity of new buses

- $nb_{rbc} \in [0, ub_{rb}] \rightarrow$ number of new b-type electric buses
- $y_{rbc} \in \{0, 1\} \rightarrow$ binary indicator defining whether b-type elec. bus is assigned to route r and charged at c-type charger
- $y_r \in \{0, 1\} \rightarrow$ binary indicator defining whether at least one elec. bus is assigned to r

Assignment of electric buses for charging

- $y_{rbc}^{(s)} \in \{0, 1\} \rightarrow$ binary indicator defining whether b-type elec. buses are charged by type c on route r according to scenario s
- $y_{bc} \in \{0, 1\} \rightarrow$ binary indicator defining whether b-type elec. buses are charged by type c on any route
- $y_{jrbc} \in \{0, 1\} \rightarrow$ binary indicator defining whether b-type elec. buses are charged by type c at non-depot stop j of route r

Variables

Charging equipment quantities

- $ns_j \in \{0, 1\}$ → binary indicator defining whether a new charger is opened at stop j
- $\alpha_{jc} \in \{0, 1\}$ → binary indicator defining whether a new charging plug device of type c is assigned to depot j
- $nc_{jc} \in [0, up_j uc_c - nod_{jc}]$ → number of new plug devices of type c at stop j
 - $nc_{jc} \in [0, uc_c - nod_{jc}], j \in D \setminus NO$
- $np_{jc} \in [0, up_j - nop_{jc}]$ → number of new charging points of type c at stop j
 - $np_{jc} \in \{0, 1\}, j \in D \setminus NO$

Allocations of power stations with charging locations

- $\beta_t \in \{0, 1\}$ → binary indicator defining whether an old or new power station is used at power station spot t
 - $\beta_t = 1; t \in TO$
- $\gamma_{tj} \in \{0, 1\}$ → binary indicator defining whether there is an old or new link between power station spot t and stop/depot j
 - $\gamma_{tj} = 1; j \in NO$ **and** $t \in TO_j$

Additional variables

- $Z_r \in [0, dem_r]$ → contribution of route r to the passenger flow
- $nv_{rb} \in [0, nv_{rb}^0]$ → number of non-battery b -type vehicles remaining on route r

Variables

- Decision variables used for the **linearization** of the equation:

$$nc_{jc} = \sum_{r \in R_{jc}} \min \left\{ \sum_{b \in B_{rc}} (nb_{rbc} + nob_{rbc}), \max \left\{ noc_{jrc}^{(ct)}, \max \left\{ \frac{ct_{rjbc} y_{jrbc}}{lt_r} \mid b \in B_{rc} \right\} \right\} \right\} - nod_{jc};$$

$$\left\{ \begin{array}{l} nc_{jrc} \in \left[0, \min \{ up_j uc_c, nc_{jrc}^{(max)} \} \right] \rightarrow \text{number of new and old plug devices of type } c \text{ used by elec. buses on route } r \text{ at stop } j \\ nc_{jrc}^{(b)} \in \left[0, \sum_{b \in B_r} (ub_{rb} + nob_{rb}) \right] \rightarrow \text{number of new and old elec. buses of type } b \text{ on route } r \text{ that charge by type } c \text{ at stop } j \\ nc_{jrc}^{(ct)} \in \left[0, nc_{jrc}^{(max)} \right] \rightarrow \text{number of new and old plug devices of type } c \text{ used by elec. buses of route } r \text{ at stop } j \end{array} \right.$$

$$+ \eta_{jrc1}, \eta_{jrc2}, \xi_{jrc}, \xi_{jrbc}$$

Constraints

- Limiting overall capital and variable cost:

$$(2) \sum_{j \in N} csta_j ns_j + \sum_{c \in C} ccp_c np_{jc} + \sum_{r \in R} \sum_{b \in B_r} (cbus_b * \sum_{c \in C_b} nb_{rbc}) + \sum_{t \in T \setminus TO} ccps_t \beta_t + \sum_{t \in T \setminus TO} \sum_{j \in N \setminus NO} cl_{tj} \gamma_{tj} \leq cc$$

$$(3) \sum_{j \in N} (vcc_j ns_j + \sum_{c \in C} vcp_c np_{jc}) + \sum_{r \in R} \sum_{b \in B_r} vcb_{rb} \leq uoc$$

- Power demand of all charging plug devices at stop j is satisfied by power station linked with j:

$$(4) \sum_{c \in C} (nc_{jc} + nod_{jc}) p_c \leq \sum_{t \in T_j} utp_t \beta_t, \quad j \in N.$$

- Any power station t is linked with **at most one** stop j:

$$(5) \beta_t = \sum_{j \in N} \gamma_{tj}, \quad t \in T.$$

- Charging type is the same for all electric buses of the same type:

$$(6) \sum_{c \in C_b} y_{rbc} - y_{rbco_b} \leq 0, \quad r \in R, \quad b \in BO \cap B_r.$$

Constraints

- B-type electric buses to be charged by a unique feasible charging type:

$$(7) \quad y_{bc} \leq \sum_{r \in R} y_{rbc}, b \in B \setminus BO, c \in C_b.$$

$$(8) \quad |R|y_{bc} \geq \sum_{r \in R} y_{rbc}, b \in B \setminus BO, c \in C_b.$$

$$(9) \quad \sum_{c \in C_b} y_{bc} \leq 1, b \in B \setminus BO$$

- Introducing upper bound on total number of old and new charging points at stop j :

$$(10) \quad \sum_{c \in C} np_{jc} + \sum_{c \in C} nop_{jc} \leq up_j, j \in N$$

- Requiring one c-type charging point to be opened at **depot $d(r)$** of route r if this route is served by at least one electric bus charged by this type:

$$(11) \quad \alpha_{jc} - np_{jc} \leq 0, j \in D \setminus NO.$$

Constraints

- Route passenger capacity coverage:

$$(12) \sum_{b \in B_r} (cap_b * \sum_{c \in C_b} nb_{rbc}) + \sum_{b \in V_r} cap_b nv_{rb} \geq dem_r^0 y_r, \quad r \in R$$

- Ensuring the number of non-battery vehicles does not increase:

$$(13) \quad nv_{rb} \leq nv_{rb}^0, \quad r \in R, b \in V_r.$$

- Ensuring selection of a single charging point location scenario:

$$(14) \quad y_{rbc} - \sum_{s=1}^{n_{rbc}} y_{rbc}^{(s)} \leq 0, \quad r \in R, \quad b \in B_r, \quad c \in C_b.$$

Constraints

- Constraints implied by definitions of the variables:

$$(15) \quad Z_r \leq \sum_{b \in B_r} cap_b \sum_{c \in C_b} nb_{rbc}, \quad r \in R$$

$$(16) \quad Z_r \leq dem_r^0 y_r, \quad r \in R.$$

$$(17) \quad nb_{rbc} \geq y_{rbc}, \quad r \in R, \quad b \in B_r, \quad c \in C_b.$$

$$(18) \quad nb_{rbc} \leq ub_{rb} y_{rbc}, \quad r \in R, \quad b \in B_r, \quad c \in C_b.$$

$$(19) \quad y_{rb} = \sum_{c \in C_b} y_{rbc}, \quad r \in R, \quad b \in B_r$$

$$(20) \quad ns_j \leq \sum_{c \in C} (np_{jc} + nop_{jc}), \quad j \in N \setminus NO$$

$$(21) \quad up_j ns_j \geq \sum_{c \in C} (np_{jc} + nop_{jc}), \quad j \in N \setminus NO$$

$$(22) \quad np_{jc} \geq \frac{nc_{jc} + nod_{jc}}{uc_c} - nop_{jc}, \quad j \in N \setminus D, \quad c \in C$$

$$(23) \quad \sum_{c \in C} np_{jc} + \sum_{c \in C} nop_{jc} \leq up_j, \quad j \in N \setminus D$$

$$(24) \quad y_r \leq \sum_{b \in B_r} \sum_{c \in C_b} y_{rbc}, \quad r \in R,$$

$$(25) \quad |B_r| y_r \geq \sum_{b \in B_r} \sum_{c \in C_b} y_{rbc}, \quad r \in R$$

$$(26) \quad \alpha_{jc} \leq \sum_{r \in R_{jc}} y_r, \quad j \in D \setminus NO, \quad c \in C$$

$$(27) \quad |R_{jc}| \alpha_{jc} \geq \sum_{r \in R_{jc}} y_r, \quad j \in D \setminus NO, \quad c \in C$$

Constraints

- Traffic interval preservation constraints:

$$(28) \quad L_r y_r \leq ut_r \sum_{b \in B_r} \left(\sum_{c \in C_b} nb_{rbc} + nob_{rb} \right) + ut_r \sum_{b \in V_r} nv_{rb}, \quad r \in R$$

$$(29) \quad L_r y_r \geq lt_r \sum_{b \in B_r} \left(\sum_{c \in C_b} nb_{rbc} + nob_{rb} \right) + lt_r \sum_{b \in V_r} nv_{rb}, \quad r \in R$$

$$(30) \quad uc_c \alpha_{jc} - nc_{jc} \leq 0, \quad j \in D \setminus NO$$

Constraints → Linearization process

$$(31) \quad nc_{jc} = \sum_{r \in R_{jc}} nc_{jrc} - nod_{jc}, j \in N \setminus D, c \in C$$

$$(32) \quad nc_{jrc}^{(b)} = \sum_{b \in B_{rc}} nb_{rbc} + nob_{rbc}, j \in N \setminus D, r \in R_{jc}, c \in C$$

$$(33) \quad nc_{jrc} \leq nc_{jrc}^{(ct)}, j \in N \setminus D, r \in R_{jc}, c \in C$$

$$(34) \quad nc_{jrc} \leq nc_{jrc}^{(b)}, j \in N \setminus D, r \in R_{jc}, c \in C$$

$$(35) \quad nc_{jrc} \geq nc_{jrc}^{(ct)} - up_j uc_c (1 - \eta_{jrc1}), j \in N \setminus D, r \in R_{jc}, c \in C$$

$$(36) \quad nc_{jrc} \geq nc_{jrc}^{(b)} - up_j uc_c (1 - \eta_{jrc2}), j \in N \setminus D, r \in R_{jc}, c \in C$$

$$(37) \quad \eta_{jrc1} + \eta_{jrc2} = 1, j \in N \setminus D, r \in R_{jc}, c \in C$$

$$noc_{jrc}^{(ct)} = \frac{\max\{ct_{rjbc} \mid b \in BO_{rc}\}}{lt_r}$$

$$nc_{jrc}^{(max)} = \left\lceil \frac{\max\{ct_{rjbc} \mid b \in B_{rc}\}}{lt_r} \right\rceil$$

Constraints → Linearization process

$$(38) \quad nc_{jrc}^{(ct)} \geq \frac{ct_{rjbc}y_{jrbc}}{lt_r}, j \in N \setminus D, r \in R_{jc}, c \in C, b \in B_{rc}$$

$$(39) \quad nc_{jrc}^{(ct)} \leq \frac{ct_{rjbc}y_{jrbc}}{lt_r} + nc_{jrc}^{(\max)}(1 - \xi_{jrbc}), j \in N \setminus D, r \in R_{jc}, c \in C, b \in B_{rc}$$

$$(40) \quad nc_{jrc}^{(ct)} \geq noc_{jrc}^{(ct)}, j \in N \setminus D, r \in R_{jc}, c \in C, b \in B_{rc}$$

$$(41) \quad nc_{jrc}^{(ct)} \leq noc_{jrc}^{(ct)} + nc_{jrc}^{(\max)}(1 - \xi_{jrc}), j \in N \setminus D, r \in R_{jc}, c \in C, b \in B_{rc}$$

$$(42) \quad \xi_{jrc} + \sum_{b \in B_{rc}} \xi_{jrbc} = 1, j \in N \setminus D, r \in R_{jc}, c \in C$$

$$nc_{jc} = \sum_{r \in R_{jc}} \min \left\{ \sum_{b \in B_{rc}} (nb_{rbc} + nob_{rbc}), \max \left\{ noc_{jrc}^{(ct)}, \max \left\{ \frac{ct_{rjbc}y_{jrbc}}{lt_r} \mid b \in B_{rc} \right\} \right\} \right\} - nod_{jc},$$

Objective function

- Linear objective function to be maximized:

$$\max \sum_{r \in R} \left(Z_r - \frac{\sum_{b \in V_r} n v_{rb} cap_b}{dem_r} \right) - \frac{\sum_{j \in N} \sum_{c \in C} n c_{jc}}{|N| \sum_{c \in C} u c_c} - \frac{\sum_{j \in N} \sum_{c \in C} n p_{jc}}{\sum_{j \in N} u p_j} - \frac{\sum_{t \in T \setminus TO} \beta_t}{|T|} - \frac{\sum_{j \in N \setminus NO} \sum_{t \in T_j} \gamma_{tj}}{|N| |T|}$$

past passenger capacity of all route r vehicles ←

passenger capacity of all buses →

upper limit on number of plug devices of one c -type charging points →

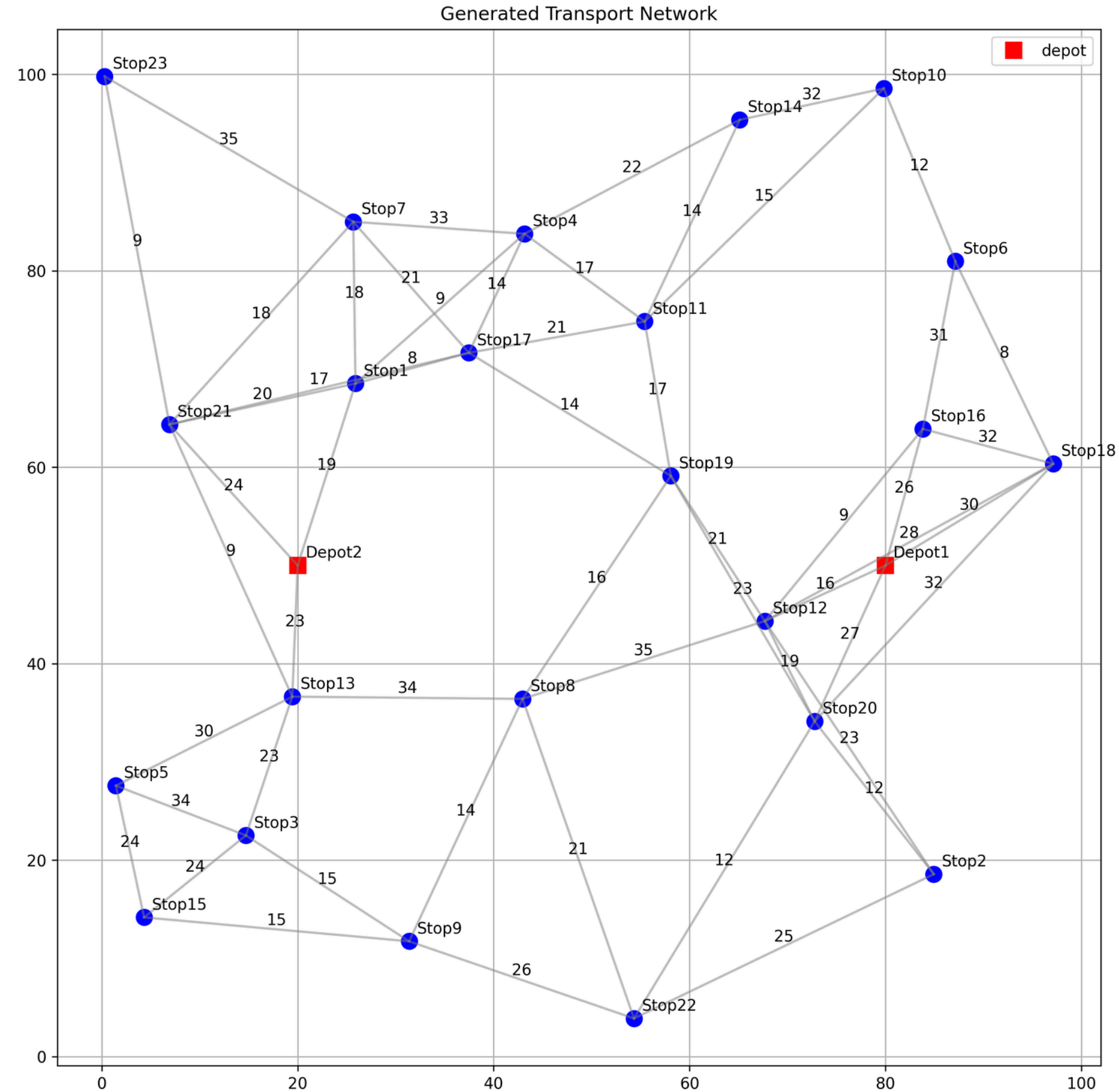
upper limit on number of charging points at stop j ←

The objective is to maximize the route-weighted total passenger capacity of new electric buses

Solution & Results

Real case instance

- **Adapted from literature:** Inspired by paper → based on real city transport system
 - # bus types → electric & conventional
 - # charging types
 - # stops & routes
 - Capacity and cost **bounds**
- **Self-constructed instance:**
 - Majority of data generated by us → randomly selected values within some range
 - Designed to **mirror realistic parameters**
- **Dynamically-created data:**
 - Graph structure (routes & stops layout)
 - Distances between stops
 - Route assignments
 - Feasible charging scenarios



Results

- **Bus types:**
 - 6 electric
 - 4 non-battery
- **Network infrastructure:**
 - 26 routes
 - 23 stops
 - 2 depots
 - 2 initial electric buses
 - 2 initial conventional buses
- **Charging:**
 - 1 charging type
 - 3 devices & plugs per stop

CHARACTERISTICS / (CC, UOC)	(40e6, 22.5e6)	(30e6, 18e6)	(16.5e6, 16e6)	(10e6, 15e6)	(10e7, 40e6)
<i>Solution time(in minutes)</i>	0.059	0.16	0.13	0.062	0.031
<i>Number of new electric buses</i>	New electric: E401: 8 units E402: 16 units E403: 24 units E404: 31 units E405: 1 unit E406: 13 units Old electric: E403: 28 units E404: 27 units Non-battery: 0 units	New electric: E401: 15 units E402: 13 units E403: 10 units E404: 32 units E405: 1 unit Old electric: E403: 27 units E404: 28 units Non-battery: M101: 11 units M102: 5 units M103: 2 unit	New electric: E401: 1 unit E402: 24 units E404: 11 units Old electric: E403: 28 units E404: 27 units Non-battery: M101: 9 units M102: 5 units M103: 1 unit M104: 1 unit	New electric: E402: 16 units Old electric: E403: 28 units E404: 27 units Non-battery: M101: 3 units M102: 2 units	New electric: E401: 15 units E402: 3 units E403: 36 units E404: 25 units E405: 4 units E406: 18 units Old electric: E403: 28 units E404: 27 units Non-battery: 0 units
<i>Total fleet size</i>	147 units	144 units	107 units	76 units	156 units
<i>Total charging stations</i>	12 units	12 units	12 units	12 units	12 units
<i>Objective value</i>	11180.485	10425.92	5785.11	2590.88	11180.496
<i>Total capital investment</i>	39.74×10^6	29.97×10^6	16.47×10^6	9.97×10^6	44.93×10^6
<i>Total variable operation costs</i>	22.45×10^6	17.99×10^6	13.46×10^6	10.75×10^6	21.95×10^6

Proposed Heuristics

5.3 ILP-based heuristics

Based on the ILP model (1)-(66) we developed two heuristics that fix some variables of the model and then solve it. At each iteration, the first heuristics HR generates a set R_{HR} of routes and sets B_r for $r \in R_{HR}$ which satisfy constraints on the capital and operating costs by removing randomly chosen buses of routes. After generation of feasible sets we fix variables $y_r = 1$ for $r \in R_{HR}$ and $y_r = 0$ for $r \in R \setminus R_{HR}$ and solve the obtained problem. The second heuristics HRBC tries to generate sets R_{HRBC} of routes, sets B_r of buses for $r \in R_{HR}$, and charger types for each bus. After generation we fix variables y_r , y_{rb} , y_{rbc} and define lower and upper bounds on nb_{rbc} in the appropriate way and solve the corresponding problem. The maximal running time for heuristics HR and HRBC was set the same as for pure ILP models (up to 5 min for groups B1, B2 and B3 and up to 10 min for groups MB1, MB2 and MB3). The maximal available time for CPLEX for solving ILP subproblems was set up to 20 sec for groups B1, B2 and B3 and up to 30 sec for groups MB1, MB2 and MB3. In Tables 9-10, we present the results obtained using these heuristics for groups B1, MB1, B2, MB2, B3 and MB3. In the Tables HR/ILP (HRBC/ILP) correspond to ratios of the value of (1) obtained by HR (HRBC) and the value of (1) obtained by solving the ILP problem. Similarly, HRBC/HR is the ratio of the values of (1) obtained by HRBC and HR.

Proposed Heuristics

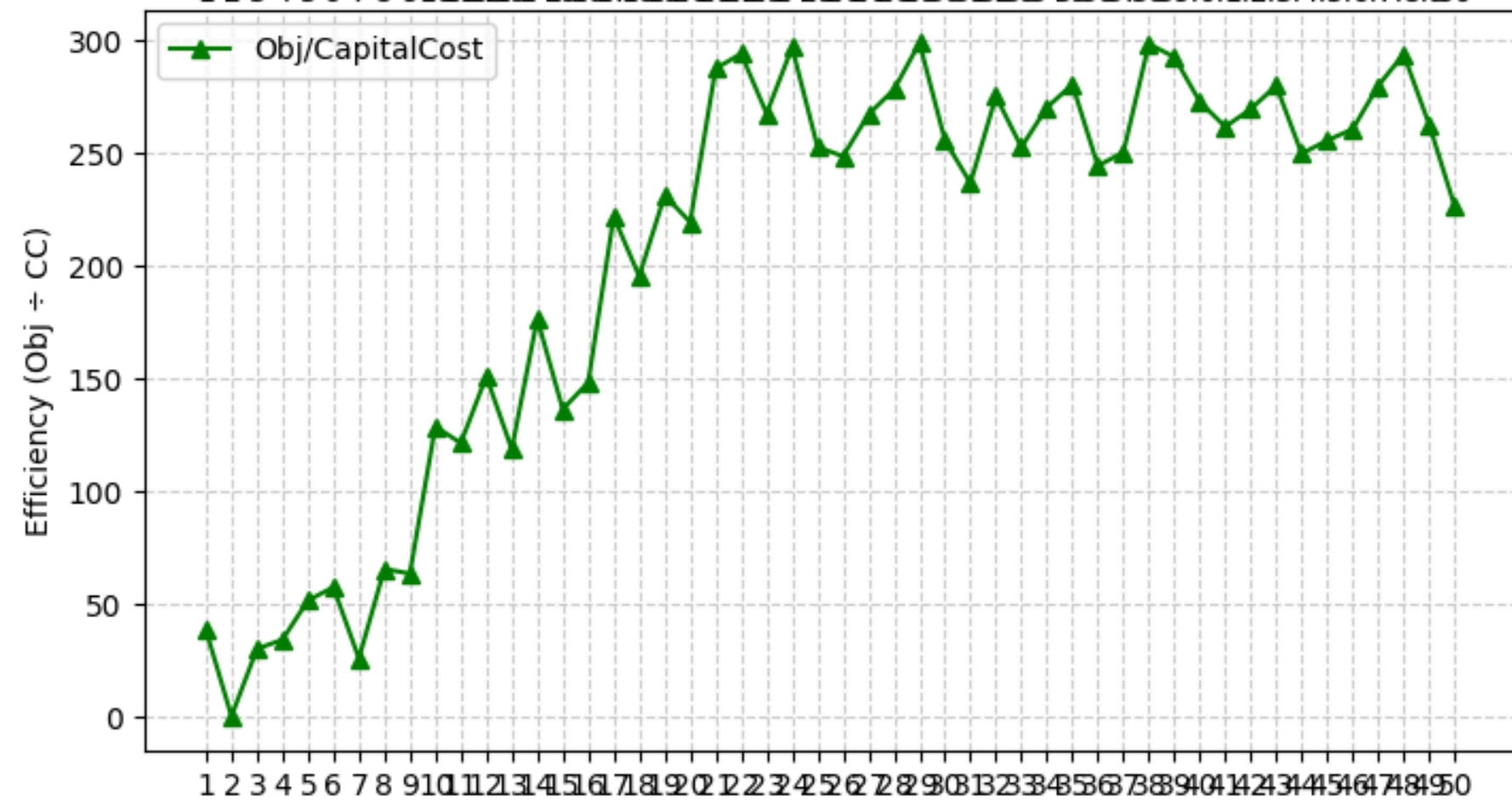
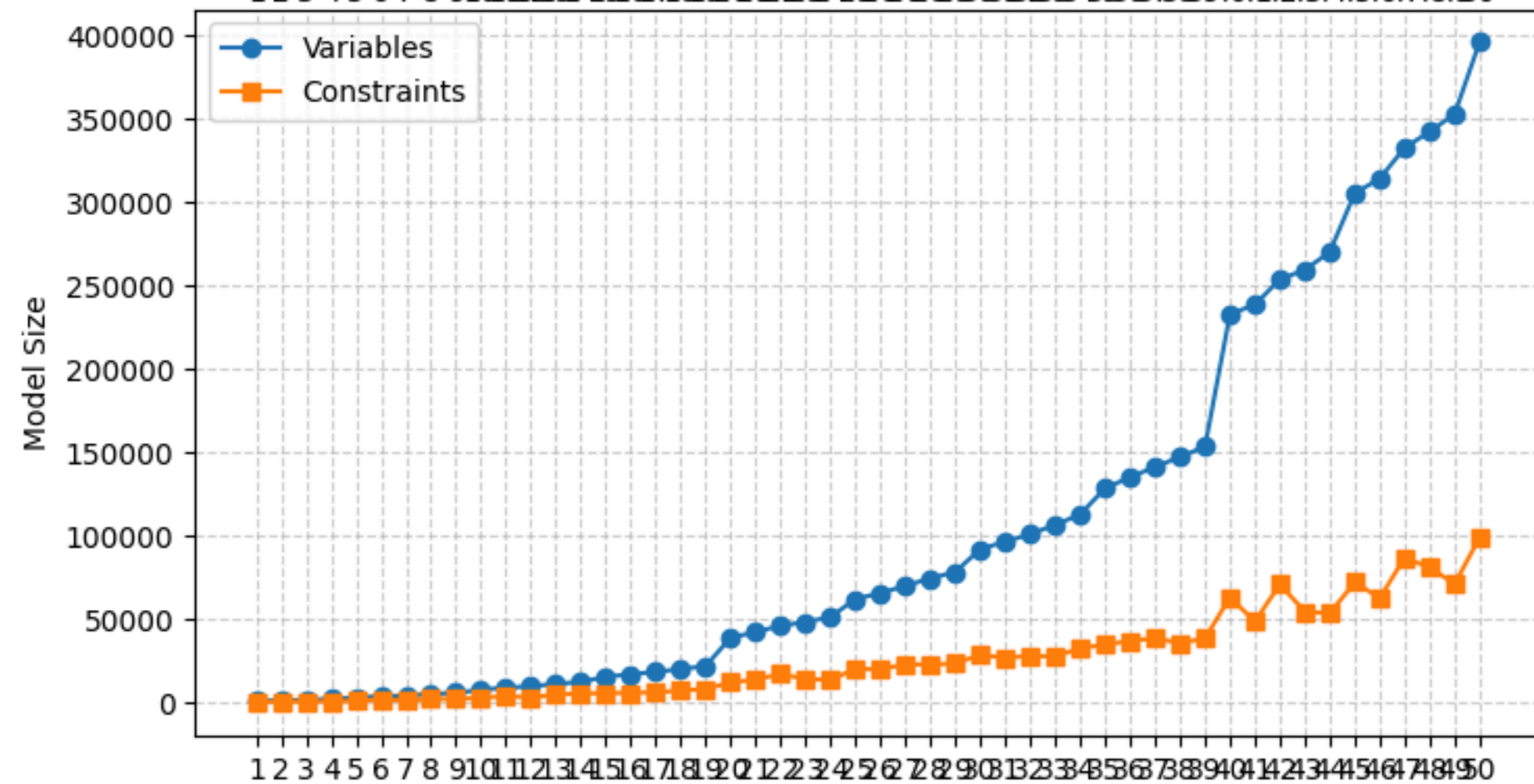
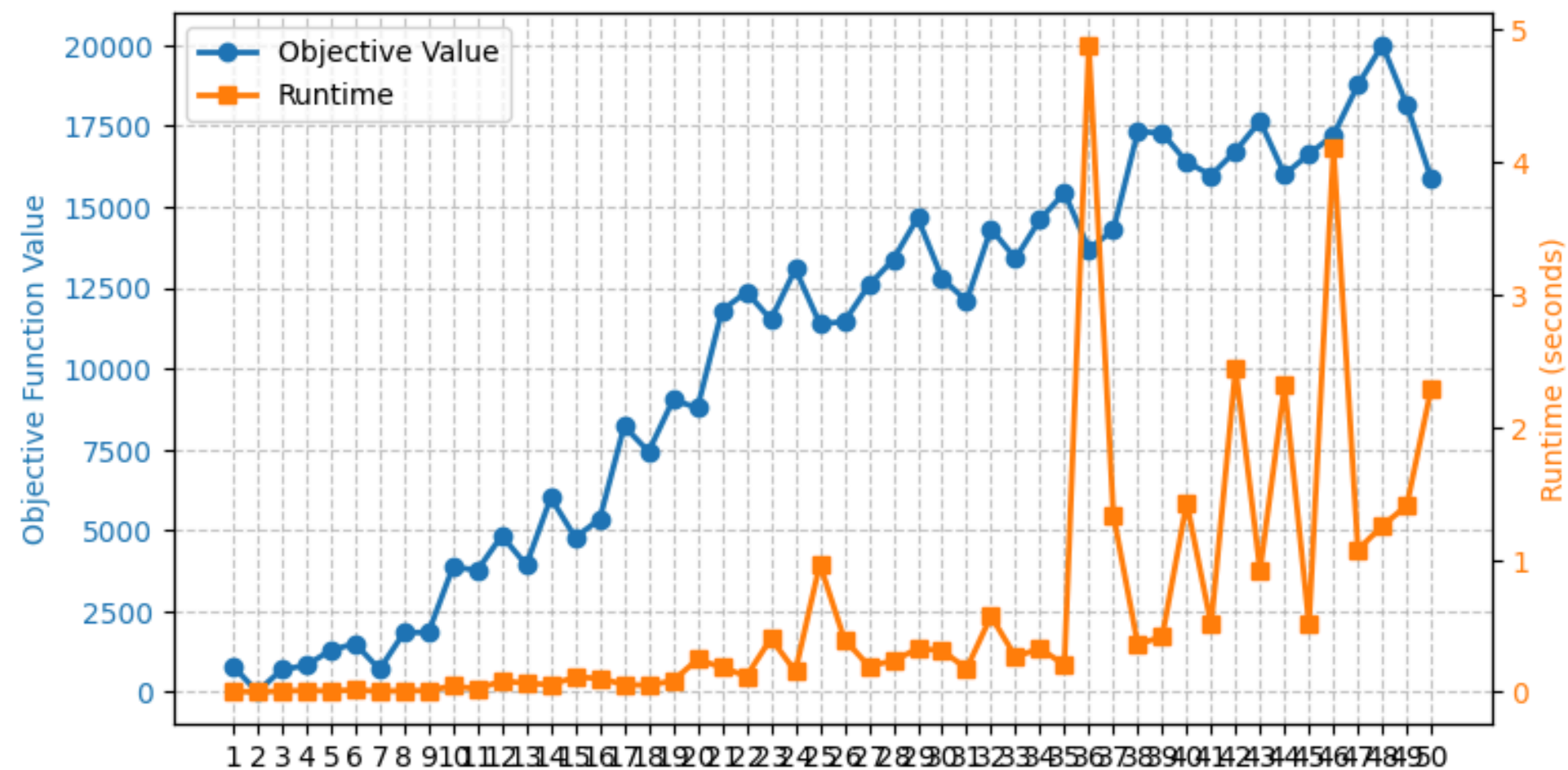
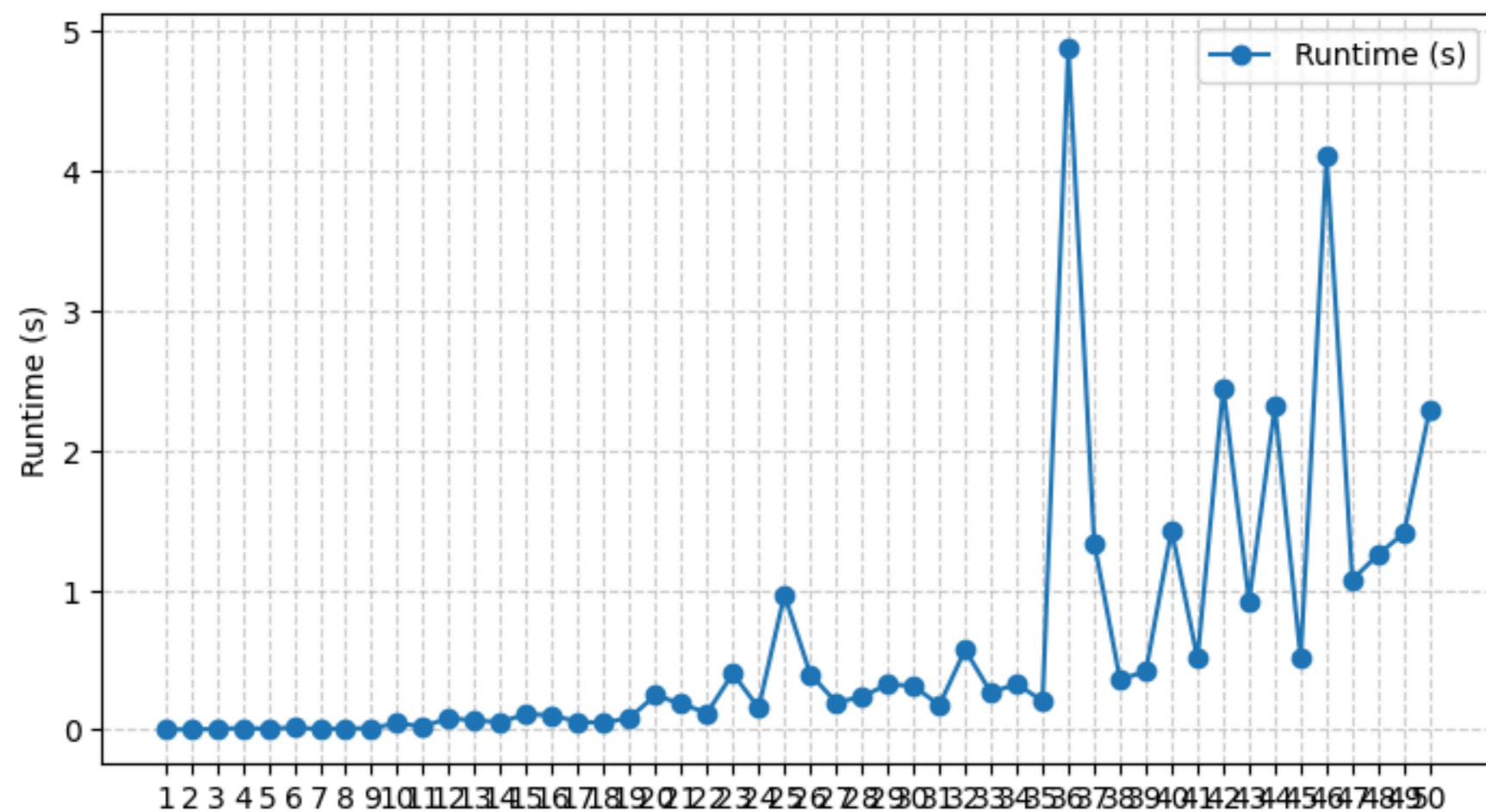
- A third heuristic mentioned in the paper
- Not described in the paper → Can be found in another parallel paper

In Table 5, we compare the results obtained with the MIP model proposed in this paper with heuristic HA presented in [7] where any stop was allowed for charging. It can be seen that heuristic HA provides a better solution for only one case out of 10 and for longer solution time.

Scalability Analysis

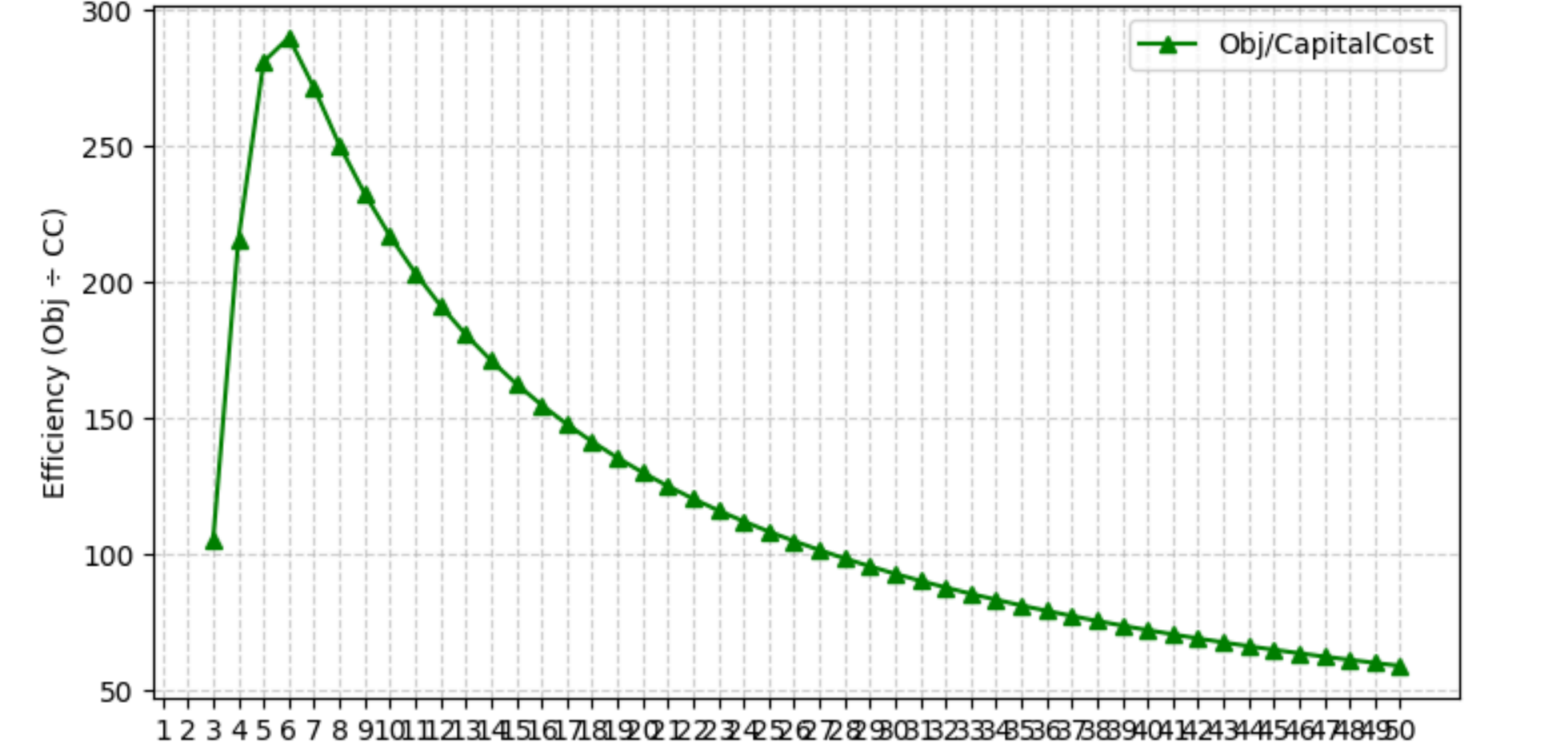
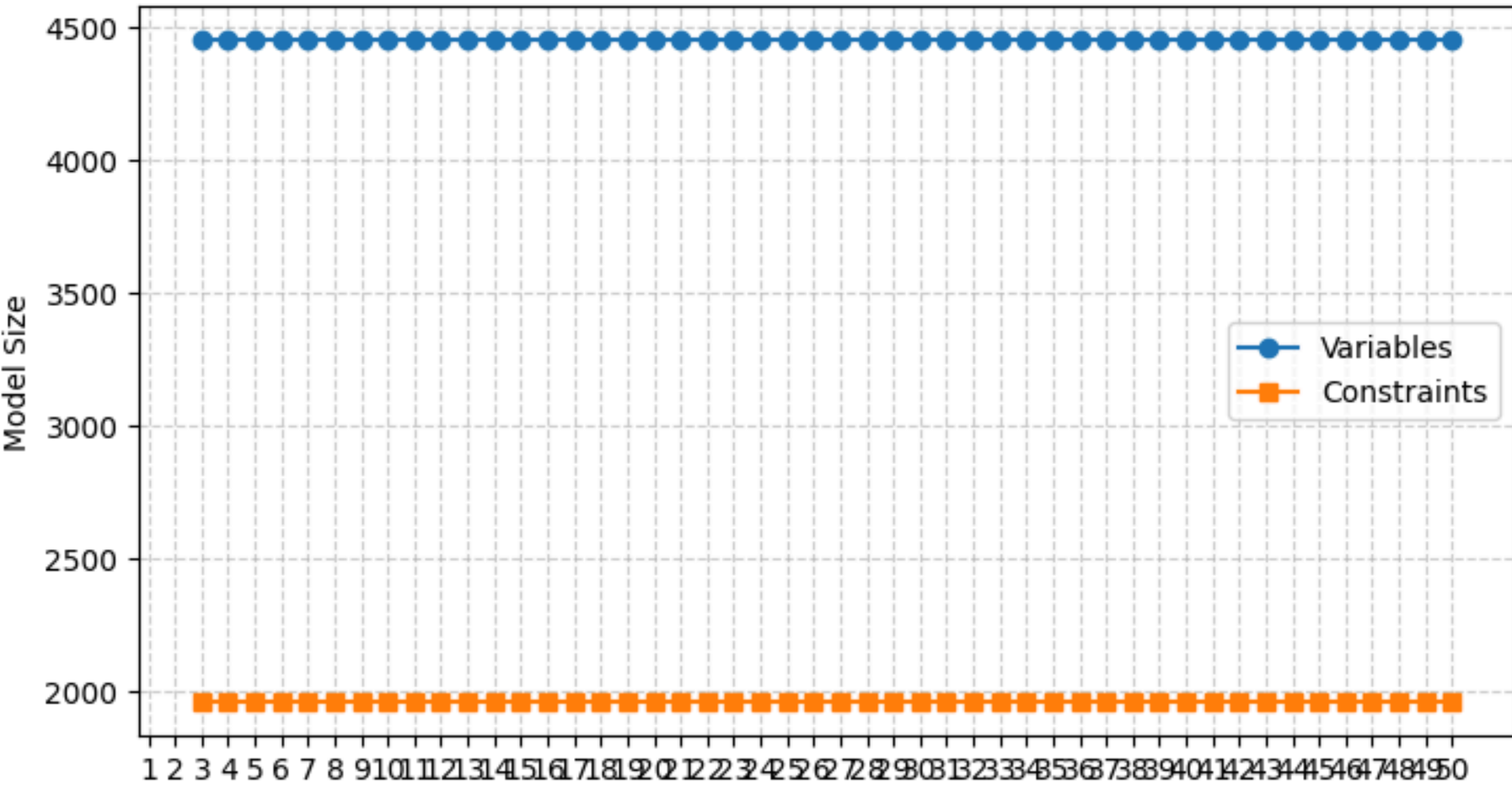
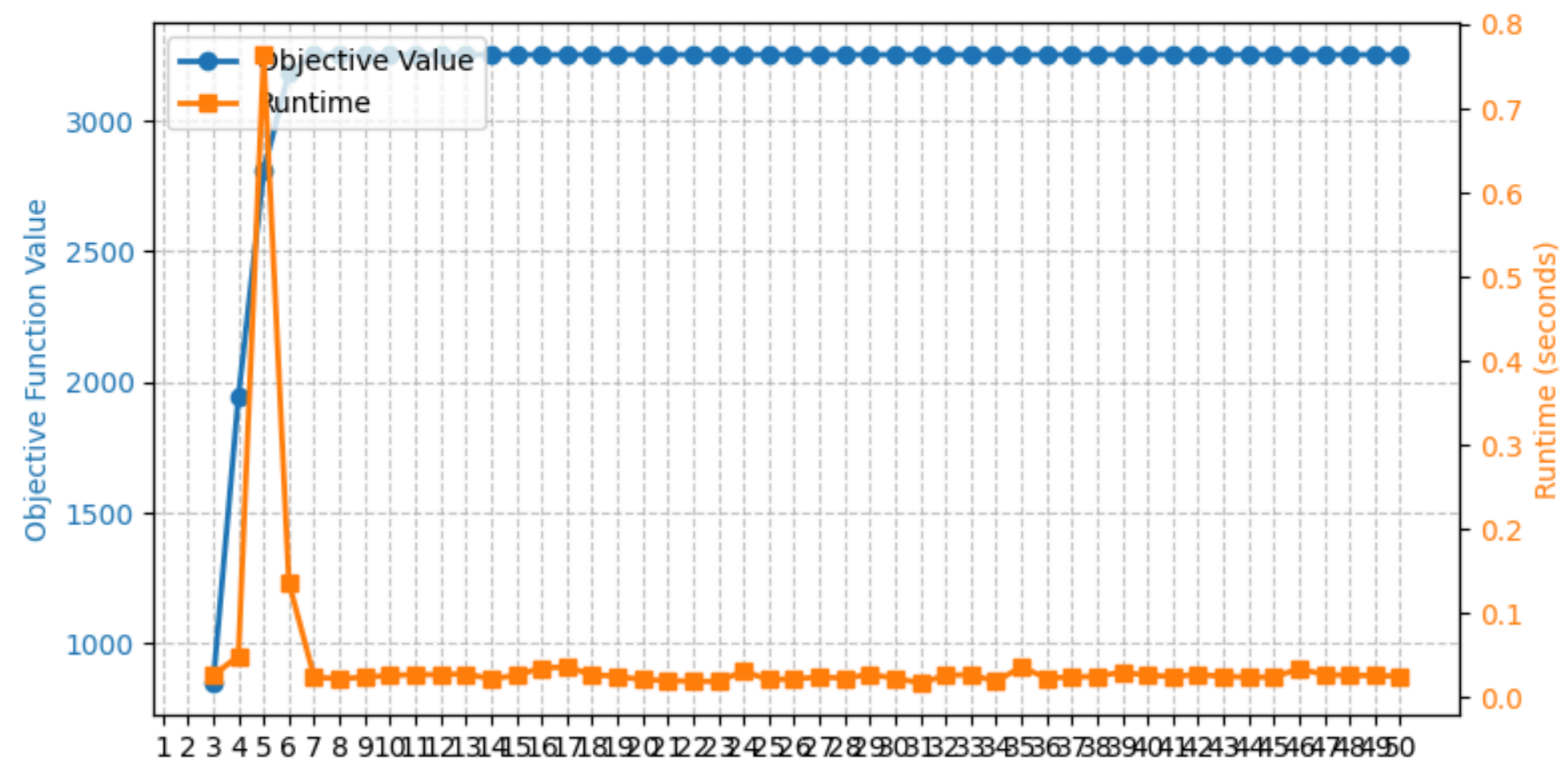
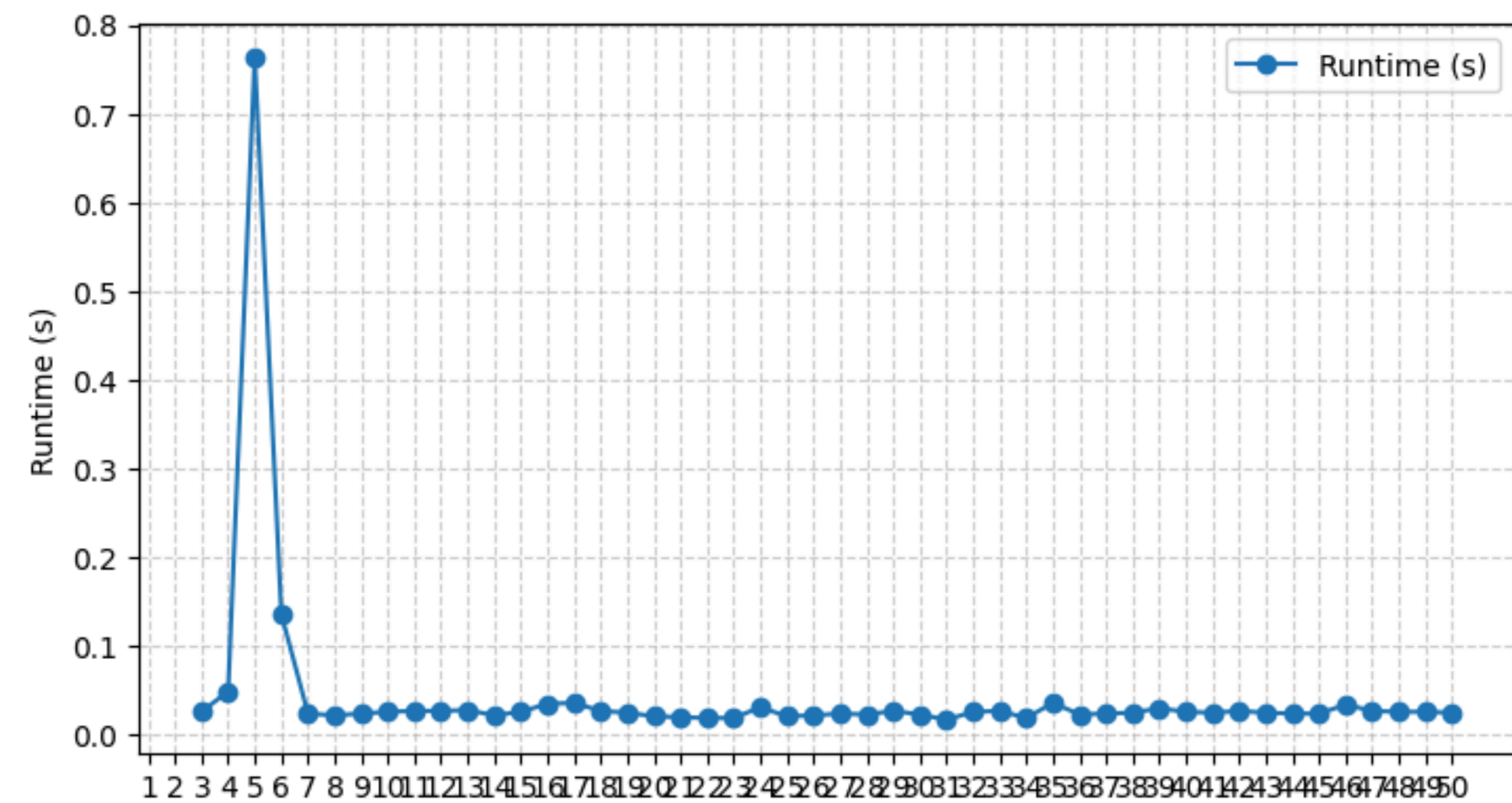
Experimental setup 1

- Understanding **runtime, memory** and **solver behaviour**
- Testing solver performance as we **increase problem size** systematically
- **Increasing parameters:**
 - Bus and Charger types → **more decision variables**
 - Routes and Stops → **more network complexity**
 - Capital cost and Operating cost → **financial scale**
- **Scaling:**
 - $\text{\#routes} = 5 + i$
 - $\text{\#stops} = 5 + i$
 - Charger/Bus type increase every few steps
 - CC and UOC increase linearly



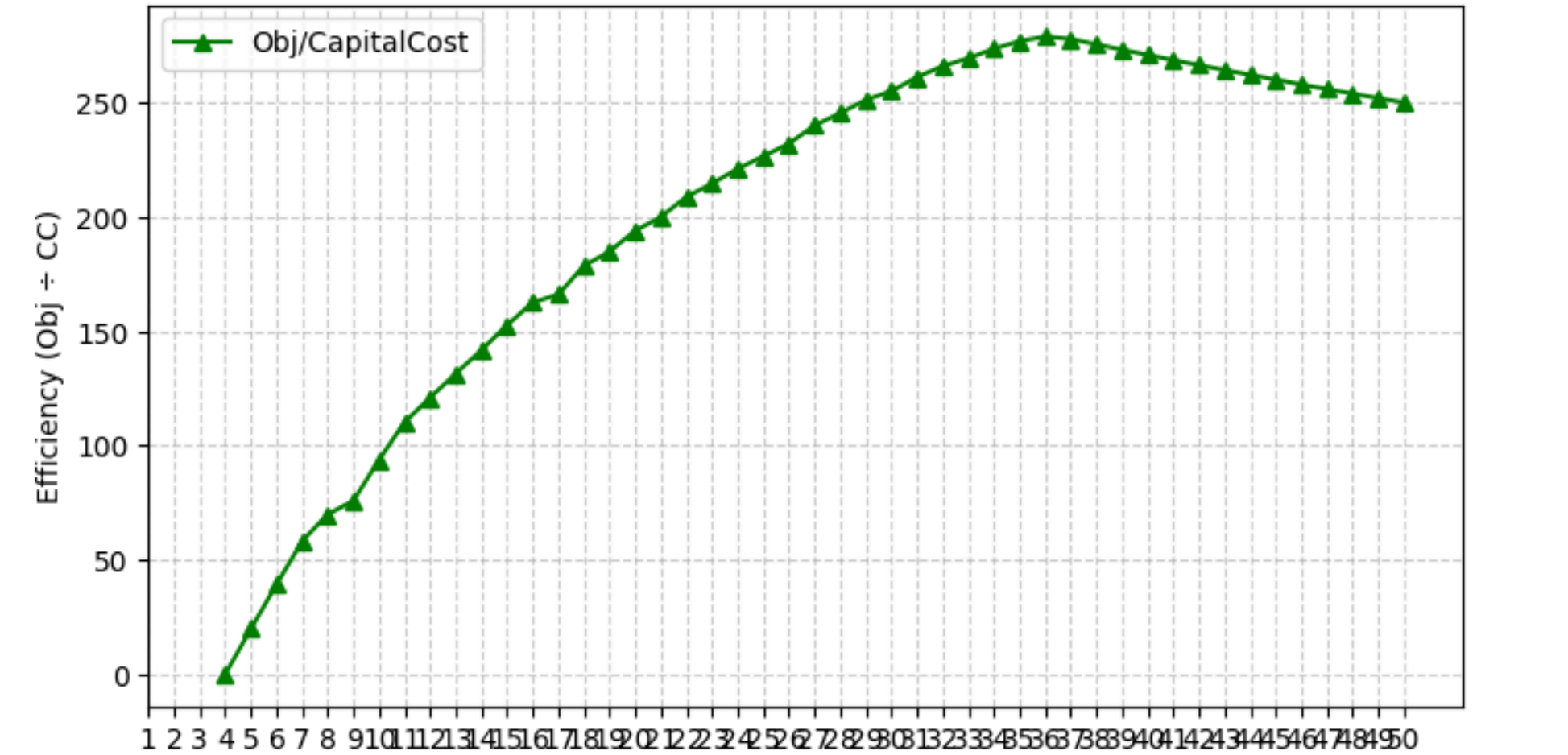
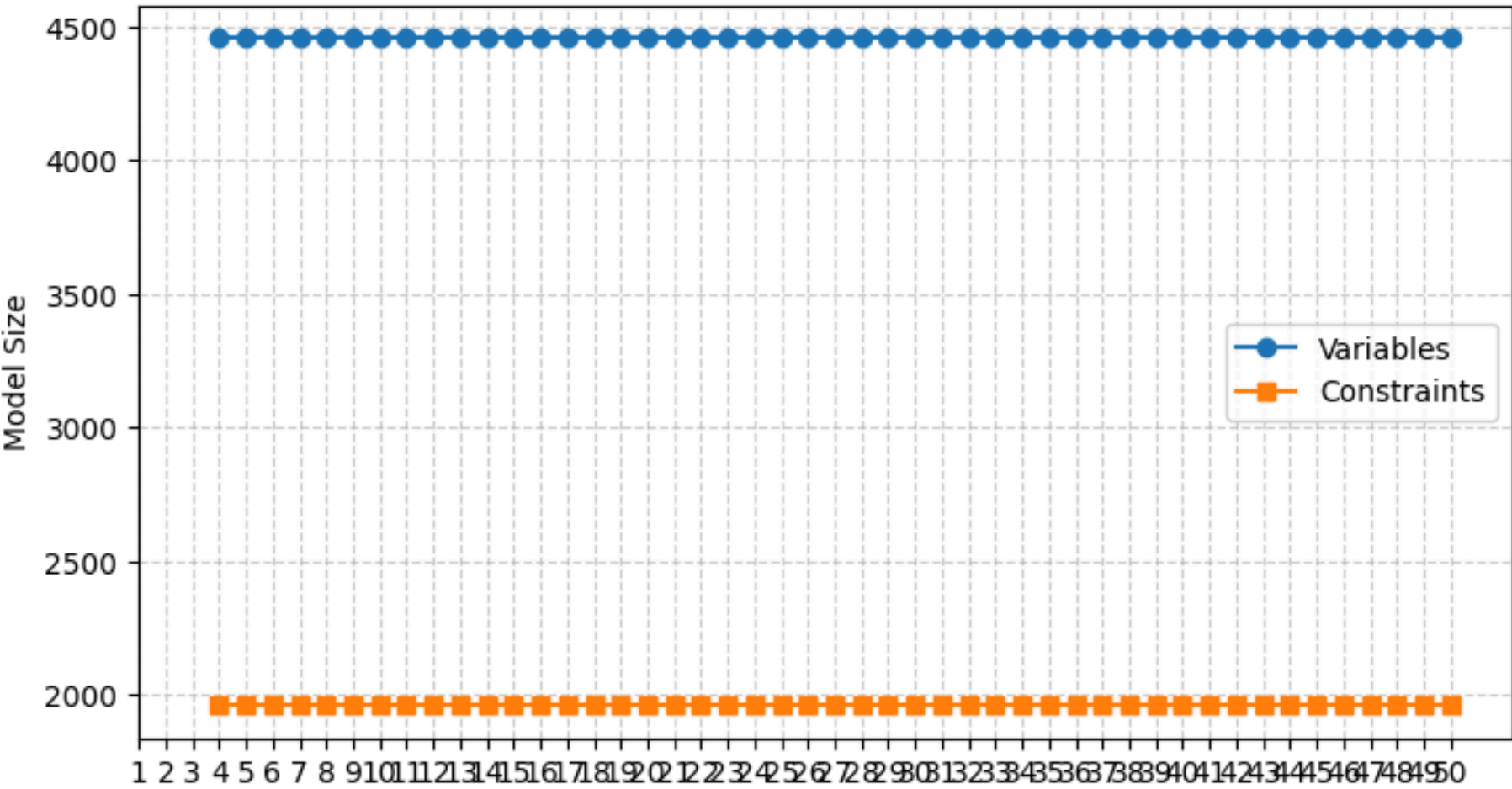
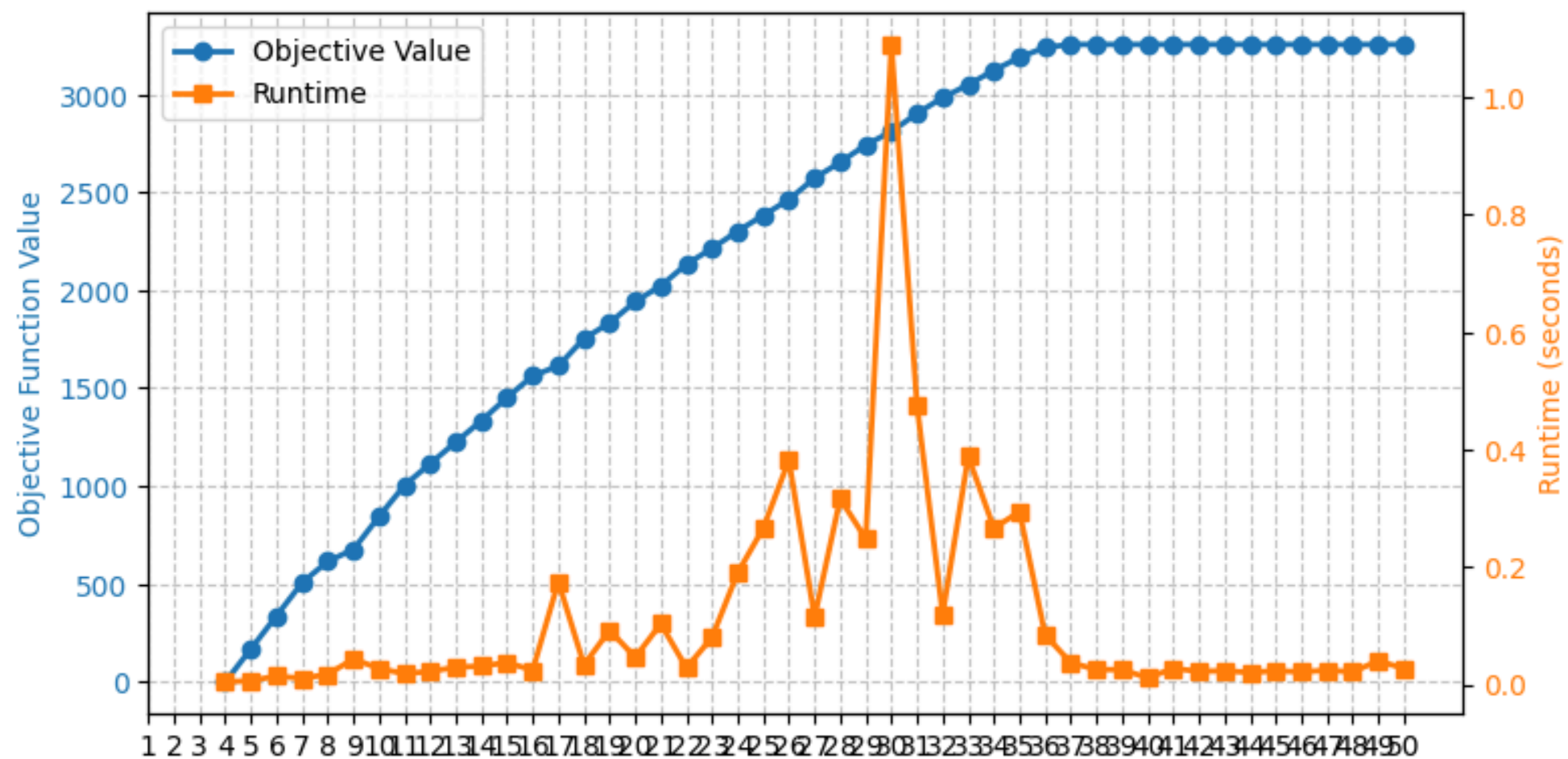
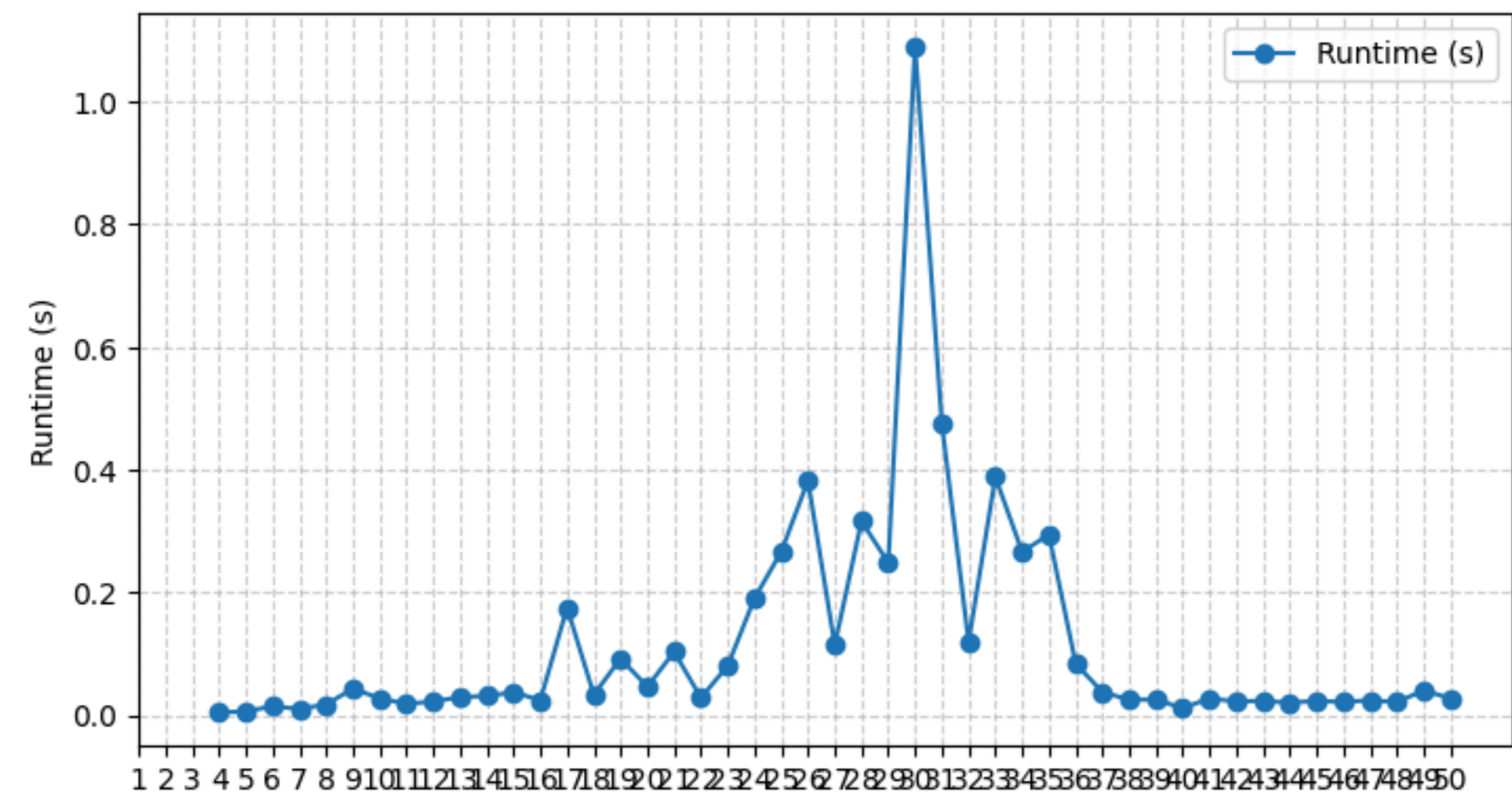
Experimental setup 2

- System structure **constant across all instances**
 - Fixed number of routes, stops, bus and charging types
 - 10 routes and 10 stops
- Only budget parameters change:
 - **CC** and **UOC** increase linearly
- Isolating the effect of **financial constraints** from system complexity
- Understanding how budget feasibility impacts solver performance



Experimental setup 3

- **Further analysis from setup 2:**
 - Low CC → infeasible
 - High CC → stable optimal solutions
 - **Transition zone** → solver runtime spikes
- Fixed system structure (routes, stops, bus/charging types)
- **Capital cost** → starts at 8M, increases by 0.1M up to 13M
- **Operating cost** → increases at the same rate



Thank you

- Further details:
 - https://github.com/loanmir/MathOpt_Paper